

Expected Time to Crack Intuition

Cracking time depends on both the **computational bottleneck** and the **probabilistic efficiency** of the guessing algorithm.

M captures the limitation or ceiling of the algorithm. For a neural network this could be a large number of generated candidates before the model stops producing useful outputs.

x_0 is the inflection point or the mid point of our function. It represents the number of guesses where success probability reaches 50%

$$\mathbb{E}[T_c] = \frac{M}{\min(\lambda, \mu)} \left(x_0 + \frac{1}{k} \right)$$

λ is the password generation speed or arrival rate. Think of this as the rate at which an ML model generates password candidates.

μ is the hash rate or service rate. It is the speed at which we generate hashes and compare our guess with the password candidate.

k is the logistic growth rate or the steepness of the curve.

High k means a sharper transition and therefore we expect more efficient candidate generation.

The intuition is that we expect our neural networks to have a lower maximum success rate M , however they offer better early performance in certain regimes due to the intelligence factor $\left(x_0 + \frac{1}{k}\right)$.

When we compare different neural networks, a larger M indicates better generalisation.

Large x_0 could mean significant difference in the distribution of the training and target data or simply the model needs to generate a large number of candidates before it start producing useful candidates.